

BST Working Paper — Volume 2: The Framework — Chapter 1: Foundations

2026-05-19 (Tuesday volume:chapter reorganization)

Section 1: The Object, and How It Is Chosen

Rewritten 2026-09-14 (Lyra, Round 144 L1; Keeper audit pending) as a pointer into the Spine plus the referee apparatus. The May 2026 text — the forced-choice chain, the five-condition elimination table, the key-results table, the one cycle — is kept below as the record. Facts are from the register (K1889, K1890, K1815, T2565, Cal Sections 946 and 959, K1813) and from `data/bst_26_tier_table_generated.md` — not from the May text. No tier table and no count is typed here: the generated table is the table, and the state block's sentence is the count. Sections 2–4 (the 2022 substrate narrative: $S^2 \times S^1$, the contact graph, the CR structure) are untouched in this pass and carry their June 2026 historical correction; whether they become record or apparatus is Keeper's call. Last accuracy-synced: 2026-09-14 / K1897.

1.0 The question

What is the simplest structure that can do physics — and how do we know we have the right one?

The first half is Lecture 1 of the Spine (Curriculum/Spine_DIV5_QM_GR_SM/Lecture_01_The_Object.md): the object is the bounded symmetric domain of type IV in complex dimension five, $D_{IV}^5 = \text{SO}_0(5, 2)/[\text{SO}(5) \times \text{SO}(2)]$, the Lie ball. The second half is Lecture 2 (Lecture_02_Why_This_Object_Honestly.md), and the answer there is more modest than this chapter once said. This section is the pointer to both and the apparatus a referee will want.

1.1 Where it stands

How the object is chosen — one leg, one integer, one input. Cartan's irreducible bounded symmetric domains — four families and two exceptions — each carry a characteristic multiplicity α : 2 for type I, 4 for type II, 1 for type III, $n-2$ for type IV_n , 6 and 8 for the exceptions. Theorem: among irreducible domains of rank at least two, $\alpha = 3$ has exactly one solution, D_{IV}^5 — verified from the standard table (K1889), rebuilt from root data across 66 dimensions with no reference consulted (Elie, toy

5750), and checked against the low-dimensional exceptional isomorphisms so nothing is counted twice (K1890-PRE). Quarks come in three colours; *the identification of that count with the characteristic multiplicity* is the program’s one dimensionless input to the choice of the object, and it is stated as an input, not derived: the geometry provably cannot supply it — the group it puts on the three-dimensional slot is $U(1) \cdot SO(3)$, dimension four, and no group anywhere in the geometry acts on that slot as colour (Elie 5751; Cal Section 946, one obstruction, ruled Section 950). The chain of “forced choices” in the May record below — observation forces $\text{rank} \geq 2$, confinement forces $N_c \geq 3$, two primality conditions, a “genus coincidence” — is retired, and it is worth saying why in one breath: four of its eleven criteria were empty, satisfied by construction (K1889); $\text{rank} = 2$ is a *posit* on the program’s own ratified floor (K1815), not a consequence of “observation”; the “genus coincidence” $n_C + \text{rank} = 2n_C - 3$ equates two integers neither of which is the genus (the genus is 5, pinned 2026-09-09); and the probability that dressed the eleven was the wrong instrument — a referee wants an enumeration, and Cartan’s classification is one. What survived is smaller and sharper than what we had.

The integers, and which kind each is. Six integers appear throughout this Guide: three are read off the classification — $\text{rank} = 2$, $n_C = 5$, $N_c = 3$ (the multiplicity a) — and three are combinations we name because they recur: $C_2 = \text{rank} \cdot N_c = 6$, $g = n_C + \text{rank} = 7 = p + q$ (the dimension of the defining representation of $SO(5, 2)$; not the genus), $N_{\max} = N_c^3 n_C + \text{rank} = 137$. Lecture 1 carries the table. Two sentences in the May text are the same input in other clothes and are not to be counted twice: “ $N_c = n_C - \text{rank}$ gives both the colours and the spatial dimensions” — the 3 of the $(3, 1)$ signature is the irreducibility of the same three-dimensional slot (T2545), so this is the colour identification read again, not a second support (Cal Section 946); and “ $N_{\max}$ matches α^{-1} ” — identified, in the bulk of a pre-registered null (K1813; Section 5). There is one dimensionful input, a mass scale (the electron mass as the ruler), and the frame of the descent is an observer input the geometry cannot select (T2565; Lecture 9). What we no longer say: “zero free parameters,” “there was never a choice,” “400+ predictions from one shape.” What we say instead: one geometry, one named dimensionless input to its choice, a short derived core, and a large number of identifications each labelled as one.

The results, and where the count lives. The May key-results table below listed twenty quantities with a precision column and no tier column. The precision column was a *match* column, and matching is what identification means; it said nothing about mechanism. The current scorecard for the 26 primary Standard-Model parameters is generated, never typed — `data/bst_26_tier_table_generated.md`, from `data/bst_26_tier_map_v2.json` via `python3 play/bst_26_tier_generator.py --emit` — with the four words (derived / identified / floored / open, plus “input”) and, for every derived row, the four-word standard on its own statement; its one-sentence count is spliced into `notes/BST_PRESENTATION_STATE_BLOCK.md` and from there into every front matter. As of this morning the count moved again (Cal Section 959 on the K962 rows; Elie’s toy 5755 sibling tests), which is exactly why it is not typed here. The reader who wants the number reads the state block; the reader who wants to know why each word is what it is reads the generated

table’s last two columns.

1.2 Apparatus — kept, labelled as apparatus

- The classification data behind the theorem: (r, a, b) per family and the two checks that make the table trustworthy — $\dim = r + ar(r-1)/2 + br$ reproduces every dimension (66 entries, zero mismatches) and $\text{genus} = (r-1)a + b + 2$ returns $d + 1$ for the balls and 5 for D_{IV}^5 (Elie 5750; the retained instruments notes/Keeper_K1889_instrument_*.py, play/toy_5750_*.py, play/toy_5751_*.py).
- The look-elsewhere price of the selector: over all 102 irreducible domains of dimension ≤ 30 and a six-item menu of invariants, “some invariant equals 3” is carried by 24 domains; the specific invariant $a = 3$ by one (Elie 5751 E2). The honest version of that price is toy 5753 (Round 142): of 48 natural (invariant, target) stories, 7 select a unique domain stably, on 6 domains, ours by one — and the control that matters, *the invariant a selects a unique domain for 25 of the 29 integer targets from 2 to 30*, so the number three is not special to the invariant; the invariant is. Had colour come in seven, the same theorem would have picked D_{IV}^9 . (Added at Keeper’s audit, 2026-09-14.) The selector’s strength rests on the invariant having been named before the target, which is a history question the register records and does not rule.
- The null behind every “key result” that is a match: K1813’s pre-registered instrument (Section 5.2): 23–26 of 76 targets score at least as well as 137; the muon-mass form scores at fewer than one in sixty (K1814). Any identification in the May table can be priced with it.
- The May footnote’s residual, as arithmetic: a 0.034% residual in m_e/m_{Pl} roughly doubles in G because $G \propto m_{Pl}^{-2}$, which is the 0.065% Lecture 9 quotes (recomputed 2026-09-11 from CODATA); the footnote’s claim that the Wyler-precision correction accounts for about 2% of the residual is the May text’s and is not re-verified here.

1.3 Tier line

- Derived (classical): $a = 3$ selects D_{IV}^5 uniquely among irreducible domains of rank ≥ 2 (K1889; toy 5750; K1890-PRE).
- Identified — the one dimensionless input to the choice of the object: colour count = characteristic multiplicity (Lecture 2).
- Proved absent: any mechanism inside the geometry for that identification (Cal Sections 946/950).
- Posit: rank = 2 on the ratified floor (K1815). Input: the mass ruler; the descent frame (T2565).
- Generated, not typed: the tier of every one of the 26 primary parameters and their count.
- Retired with this section: the five-condition elimination and its table; the “genus coincidence”; “zero free parameters”; “400+ predictions”; the CMB-wreckage reading (K1813’s class: quote-anything, target not evidence).

- Not claimed: forcing; zero free parameters; “the unique geometry that can do physics”; any probability.

What would make this section wrong: the theorem fails only if Cartan’s table is wrong, and you can rebuild it (toy 5750); the identification fails if colour is not three, and it is; the honest exposure is the price — every later result that leans on $N_c = 3$ inherits one measured input.

1.4 May 2026 record

Below: the May 2026 text, with the in-place dated corrections of 2026-09-11 already in it (the 1.1 sentence on “zero free parameters,” the 1.3 sentence on g and the genus, the 1.6 table’s Cabibbo row and its head-note). Read with 1.1–1.3 in hand: its Steps 3–4 and the elimination table are the retired uniqueness argument; its “five invariants locked by the genus coincidence” is retired; its key-results table is a match table, and the register’s word for every row is in the generated table; its particle table (1.7) is the narrative reading of the 2022 substrate picture, kept as the record of what we once wrote.

May 1.1 — The Question What is the simplest geometric object that can produce a universe with observers?

This paper answers that question. The answer is D_{IV}^5 , the type IV bounded symmetric domain of complex dimension 5 in Cartan’s classification. From this single geometry, with one named input (the colour identification), BST reads a short derived core and several hundred identifications across 66 physical domains — from the fine structure constant (identified) to the genetic code (identified). Every derived claim names its mechanism; every identification is labelled as one. (This sentence said “zero free parameters” and “derives 400+ predictions” until 2026-09-11.)

The argument proceeds through a chain of forced choices. At each step, the reader has no alternative but to follow. There is no branching, no selection, no landscape.

May 1.2 — The Chain of Forced Choices Step 1: It must be a geometry. Physics happens in space. Space IS geometry. Any theory of everything must begin with a shape, not an equation. Equations describe shapes; shapes exist independently of the equations we write for them.

Step 2: The menu is finite and known. Élie Cartan classified all bounded symmetric domains in 1935. There are exactly four infinite families (Types I–IV) and two exceptional cases (E_6, E_7). This classification is complete — there is no undiscovered Type V.

Step 3: Apply minimum requirements. Any geometry that produces a physical universe with observers must satisfy:

1. Observation ($\text{rank} \geq 2$): Self-referential measurement requires two independent spectral directions for triangulation. A rank-1 system propagates but

cannot self-locate (T317, T944).

2. Confinement ($N_c \geq 3$): Quarks must bind. Asymptotic freedom requires $N_c \geq 3$. Without confinement, no stable matter forms.
3. Spectral integrity (g prime): The Bergman genus must be prime, or the spectral structure factorizes into sub-lattices that pull apart.
4. Channel integrity ($N_{\max}$ prime): The spectral ceiling must be prime. A composite $N_{\max}$ decomposes the fine structure constant, creating sub-channels that interfere destructively.
5. Internal consistency (genus coincidence): The embedding dimension $g = n_C + \text{rank}$ must equal the topological genus $g = 2n_C - 3$. If these two independent geometric quantities disagree, the domain cannot sustain a self-consistent spectral theory.

Step 4: One survivor (T953). Apply these five conditions to every entry in Cartan's classification:

Domain	Rank	N_c	g	$N_{\max}$	Obs.	Conf.	g prime	$N_{\max}$ prime	Genus	Verdict
D_{IV}^3	2	1	5	7	✓	×	✓	✓	×	Dead
D_{IV}^4	2	2	6	34	✓	×	×	×	×	Dead
D_{IV}^5	2	3	7	137	✓	✓	✓	✓	✓	Survives
D_{IV}^6	2	4	8	386	✓	✓	×	×	×	Dead
D_{IV}^7	2	5	9	877	✓	✓	×	✓	×	Dead
$D_I^{p,q}$	$\min(p,q)$	—	—	var.	—	—	—	—	×	Dead
D_{II}^n	$\lfloor n/2 \rfloor$	var.	—	var.	—	—	—	—	×	Dead
D_{III}^n	n	var.	—	var.	—	—	—	—	×	Dead
E_6	—	14	—	—	—	—	—	—	×	Dead
E_7	3	—	—	—	×	—	—	—	—	Dead

*Rank 3 violates the depth ceiling (T421, T944): 1135/1135 surveyed theorems have depth ≤ 1 , requiring rank ≤ 2 .

D_{IV}^5 is the unique geometry that satisfies all five conditions. This is not selection — it is elimination. There was never a choice.

Step 5: Read the invariants. D_{IV}^5 has exactly five invariants, all determined by its root system and spectral structure (Section 1.3). These are not parameters to be measured. They are the geometry's measurements of itself.

Step 6: Derive the physics. Those five invariants produce 400+ predictions across 66 domains (Section 43). All from one shape.

Step 7: Check the wreckage. If competing geometries briefly existed at the Big Bang and collapsed, their remnants should be visible. The CMB anomalies — low quadrupole, cold spot, parity asymmetry, hemispherical asymmetry — all cluster at multipoles $\ell < 30$, matching the failed manifolds' integer values (Section 46.16,

T953, Toy 1000). The six known CMB anomalies correspond to six failed geometries.

May 1.3 — The Five Invariants: One Geometry, Five Readings The domain $D_{IV}^5 = \text{SO}_0(5, 2)/[\text{SO}(5) \times \text{SO}(2)]$ is specified by two irreducible quantities: rank = 2 (from the type IV root system) and $n_C = 5$ (the complex dimension). These are locked by the genus coincidence: $n_C + \text{rank} = 2n_C - 3$ has the unique solution $(n_C, \text{rank}) = (5, 2)$. All remaining integers follow by arithmetic:

$$N_c = n_C - \text{rank} = 3, \quad g = n_C + \text{rank} = 7, \quad C_2 = \text{rank} \times N_c = 6$$

$$N_{\max} = N_c^3 \times n_C + \text{rank} = 135 + 2 = 137$$

The five integers are five readings of one object. $N_c = 3$ gives both the number of quark colors and the number of spatial dimensions — same number, same reason: it is what remains after the observer claims its rank. The integer $g = 7 = p + q$ — the defining-representation dimension of $\text{SO}(5, 2)$, not the genus, which is 5 — sets the spectral ceiling. The channel capacity $N_{\max} = 137$ is a computed invariant that matches α^{-1} ; the match is identified, not derived (Spine Lecture 8).

Restatement: BST does not begin with five integers. BST begins with one geometry — the unique bounded symmetric domain that supports self-referential observation — and reads five integers off it. Everything that follows is consequence.

What is the simplest object that can do physics? Among all bounded symmetric domains in Cartan’s classification, D_{IV}^5 is the one whose rank agrees with its dimension, whose colors agree with its space, whose arithmetic agrees with its physics, and whose observers agree with its structure. Every other geometry has an internal contradiction — a part that fights another part. D_{IV}^5 has none.

The universe is the geometry that cooperates best.

May 1.4 — The Previous Framing The original 2022 BST proposal describes the universe as the three-dimensional projection of a substrate communicating through a one-dimensional channel. The substrate geometry was originally written as $S^2 \times S^1$ and described as “circles tiling a sphere.” Per the June 2026 historical correction (see README), the dimension count in this original framing was wrong: the substrate is properly the 5-dimensional Shilov boundary $\partial_S D_{IV}^5 = (S^4 \times S^1)/\mathbb{Z}_2$ of the bounded symmetric domain D_{IV}^5 , with the configuration-space bulk being the holomorphic extension of substrate boundary data via the Hardy decomposition. The substantive structural logic of Section 2 (codimension-1 boundary structure with phase communication, derived from minimality) remains valid; the dimension count is what needed correction. The forced-choice argument of Section 1.2 provides a complementary, geometry-first route to the same destination.

On the manifold D_{IV}^5 , mathematics and physics are unified. The Laplacian eigenvalue that determines the mass gap is the same eigenvalue that determines the spectral zeta function is the same eigenvalue that determines the location of the Riemann zeros. The Selberg trace formula is the partition function. Arthur parameters are excitation modes. The functional equation is unitarity. These are not analogies — they are identities, consequences of the single underlying geometry. The 19 free parameters of the Standard Model are not inputs the universe requires; they are arithmetic complexity — the overhead introduced by methods that do not know they are computing on D_{IV}^5 .

A note on cooperation. The word “cooperation” has technical status in BST. When we say D_{IV}^5 is the geometry that cooperates best, we mean something precise: it is the unique bounded symmetric domain whose rank, dimension, colors, genus, and channel capacity are mutually consistent — where no invariant contradicts another. In every other Cartan domain, at least one structural requirement fights the others (Section 1.2, Table). Cooperation in BST is not a social metaphor applied to physics. It is a geometric property: the absence of internal contradiction among independently determined invariants. The cooperation threshold $f_{\text{crit}} = 20.6\%$ (T678), the cooperation-defection phase transition, and the emergence of stable matter all trace to this same structural fact. The universe does not merely permit cooperation — cooperation is what its geometry IS.

May 1.5 — Scope of This Paper This paper presents the complete BST framework in 46 sections, from the forced-choice derivation of D_{IV}^5 (Section 1) through substrate geometry (Section 2), configuration space and physical constants (Section 3–6), forces and nuclear physics (Section 7–8), relativity, gravity, cosmology, dark matter, weak force, thermodynamics, antimatter, and the growing manifold (Section 9–24), broader implications (Section 25–28), the deep mathematical structure connecting D_{IV}^5 to the Riemann zeta function (Section 29–31), the Riemann Hypothesis proof (Section 32–35), 27 uniqueness conditions (Section 37.5), Arithmetic Complexity, Navier-Stokes, BSD, Hodge, Four-Color, Fermat/Poincaré, and Unification (Section 36–42), 400+ experimental predictions (Section 43), the research program and cosmological cycles (Section 44–45), and recent results including science engineering, spectral-arithmetic closure, sector assignment, and manifold competition (Section 46).

May 1.6 — Key Results at a Glance All results below are read off the geometry of D_{IV}^5 with one named input; the tier column, where present, says which are derived and which identified, and the register wins where this table is stale. Precision is relative to CODATA measured values.

Quantity	BST Formula	Precision	Section
Fine structure constant α^{-1}	$\rho_2^2 (\text{Vol } D_{IV}^5)^{1/4} / (2\pi^4) 0.0001\%$ — HC Weyl vector $\rho_2 = (n_C - 2)/2$		5.1

Quantity	BST Formula	Precision	Section
Muon/electron mass ratio m_μ/m_e	$(24/\pi^2)^6$ — Bergman kernel ratio to the 6th power	0.003%	7.5
Proton/electron mass ratio m_p/m_e	$(n_C+1)\pi^{n_C} = 6\pi^5$	0.002%	7.4
Cosmological constant Λ	$g \cdot e^{-C_2(g^2-\text{rank})} = 7e^{-282}$ (T1485)	0.076 dex	12.5
Phase transition temperature T_c	$N_{\max} \times 20/21$	0.018%	15.1
Gravitational constant G	$\hbar c (6\pi^5)^2 \alpha^{24}/m_e^2$ — $12=2C_2$ Bergman round trips	0.07%	10.3
Contact scale d_0/ℓ_{Pl}	$\alpha^{14} \times e^{-1/2}$	—	12.5
Strong coupling $\alpha_s(m_p)$	$(n_C+2)/(4n_C) = 7/20$	~0%	7.6 note
Weinberg angle $\sin^2 \theta_W$	$N_c/(N_c+2n_C) = 3/13$	0.2%	6.3
Baryon asymmetry η_b	$(3/14)\alpha^4 = N_c/(2g) \times \alpha^4$ (T929)	0.45%	7.6 note
Hubble constant H_0	67.29 km/s/Mpc (Route C: full CAMB, Toy 677)	0.1%	12.6
Neutrino masses m_{ν_2}, m_{ν_3}	$(7/12)\alpha^2 m_e^2/m_p$, $(10/3)\alpha^2 m_e^2/m_p$	0.35%, 1.8%	7.6
PMNS angles $\theta_{12}, \theta_{23}, \theta_{13}$	$(3/10)(44/45)$, $(4/7)(44/45)$, 1/45 (T1446)	0.06%, 0.40%, 0.9%	7.7
Cabibbo angle $\sin \theta_C$	$1/\sqrt{20}$ (T2530, DERIVED blind 2026-07-29; the May form $2/\sqrt{79}$ /T1444 is retired)	-2.06σ on the direct value; exactly first-row unitary	7.7
CKM CP phase γ	$\arctan(\sqrt{n_C}) = \arctan(\sqrt{5})$	0.6%	7.7
Jarlskog invariant J_{CKM}	$\sqrt{2}/50000$	2.1%	7.7

Quantity	BST Formula	Precision	Section
Fermi scale v (Higgs vev)	$m_p^2/(g \cdot m_e) =$ $36\pi^{10}m_e/7,$ $g = 7 = p + q$ (not the genus)	0.046% (identified)	14.7
W boson mass m_W (Route B)	$n_C m_p/(8\alpha)$	0.02%	14.7
SPARC rotation curves (175 galaxies)	Channel noise, no dark matter	$\chi^2/\nu < 1$	19
NANOGrav GW spectrum	Phase transition at $T_c = 0.487$ MeV, $f_{\text{peak}} \approx 6\text{--}9$ nHz	In band	15.6

†The 0.034% residual in m_e/m_{pl} (and hence G) has no clean closed-form identification. The Wyler formula precision ($\Delta\alpha/\alpha \approx 6 \times 10^{-7}$, amplified $12\times = 0.0007\%$) accounts for only $\sim 2\%$ of it. No simple one-loop QED formula matches. The residual $\Delta S = 0.000326$ in the Bergman action is an open calculation.

May 1.7 — The One Cycle The universe runs one essential cycle:

Light is emitted $\rightarrow$ touches the universe $\rightarrow$ brings back information $\rightarrow$
information is stored $\rightarrow$ the substrate emits light $\rightarrow$ the cycle continues.

This is the literal operation of the contact graph on $S^2 \times S^1$. A committed contact releases a phase oscillation (photon). The oscillation reaches another bubble. Phases compare — information is exchanged. The receiving bubble commits — its phase is permanently determined, the contact graph grows by one entry. The newly committed contact participates in the next round. Every physical phenomenon is this cycle running on $S^2 \times S^1$ with configuration space D_{IV}^5 .

Every particle plays a role in this cycle:

Particle	What it IS on the substrate	Role in the cycle
Photon	Phase oscillation on S^1 , zero winding	The messenger — carries information between contacts
Electron	One complete S^1 winding — the minimal circuit	The simplest persistent commitment
Proton	Three quarks, Z_3 closure on $\mathbb{CP}^2$ — first bulk resonance	The first complete sentence — mass gap = $6\pi^5 m_e$

Particle	What it IS on the substrate	Role in the cycle
Neutron	Proton with one flavor changed via Hopf intersection	The proton rephrased — same structure, different content. Decays into the three fundamental roles: proton (matter), electron (connection), neutrino (vacuum)
Neutrino	The vacuum quantum — propagating ground state of D_{IV}^5	The silence between words — the vacuum checking on itself
Quarks	Partial Z_3 circuits on $\mathbb{CP}^2$ — fragments	Letters, not words — meaningful only in combination
W/Z bosons	Hopf fibration excitations on $S^3 \rightarrow S^2$	The editor — changes meaning (flavor) at a cost (mass)
Gluon	Z_3 phase mediator on $\mathbb{CP}^2$	The binding — holds the sentence (baryon) together
Dark matter	Channel noise — incomplete windings, non-integer	The blank page — not empty, has capacity
Higgs	Scalar fluctuation of committed Hopf geometry	The choice — which vacuum orientation was committed

The electron (boundary excitation, $k = 1$, below Wallach set) is light because it lives on the Shilov boundary $\check{S} = S^4 \times S^1$. The proton (bulk resonance, $k = 6$, Bergman space π_6) is heavy because it resonates through the full interior of D_{IV}^5 . The neutrino ($m_1 = 0$ exactly) is the vacuum itself — the substrate's geometric ground state in propagating form. The mass ratio $m_p/m_e = 6\pi^5$ measures how much more geometry the proton traverses compared to the electron's single winding.

Full particle descriptions: `notes/BST_ParticleFamily_Portrait.md`.

Section 2: The Minimum Structure

Keeper's ruling, 2026-09-14 (Lyra asked): Sections 2–4 — the 2022 substrate narrative ($S^2 \times S^1$, the contact graph, the CR structure) — are the May 2026 record, kept as motivation and history, not as the definition of the object. The object is D_{IV}^5 as one operator (Lecture 1); the genesis story is how it was first recognised (the July scoping's

own rule: *genesis is motivation, not definition*). *Nothing below carries a tier; read Section 1 first.*

2.1 Deriving $S^2 \times S^1$ from First Principles

BST's substrate geometry is not chosen from a menu of possibilities. It is the unique answer to a single question: *what is the minimum structure capable of producing physics?*

The derivation proceeds in four forced steps, each necessitated by the inadequacy of the previous answer.

Step 1: The simplest object. Begin with nothing and ask what the simplest possible structure is. A one-dimensional object — a line. But an open line has endpoints. Endpoints are boundaries. Boundaries require boundary conditions, which require additional structure to specify. An object with boundaries is not minimal because the boundaries themselves need explanation.

Step 2: The simplest closed object. The simplest closed one-dimensional object is a circle — S^1 . It has no boundary, no endpoints, no edge conditions to specify. It is completely self-contained. The circle is the minimum self-sufficient one-dimensional structure.

Step 3: Interaction requires tiling. A single circle is isolated and cannot interact with anything. Multiple circles can interact by touching — sharing contact at their edges. Circles tiling a surface account for interaction through contact. The simplest closed surface that can be tiled by circles is the sphere — S^2 . The tiling is the contact graph. The contact points are where physics happens. A single line cannot tile a surface; a circle tiles by packing, creating rich contact geometry.

Step 4: Communication requires a channel. Circles in contact on a surface are static without a means of communication. Each circle already has a natural communication degree of freedom: its phase. A position on S^1 parameterizes the relationship between each pair of contacting circles. This phase provides the third dimension — not as additional space but as the channel through which tiled circles exchange information.

2.2 Uniqueness

Each step is forced by the failure of the simpler alternative. The result — circles tiling a sphere, communicating through their shared circular phase — is $S^2 \times S^1$. This is the unique minimum structure that is closed (no boundaries), interacting (contact graph), and dynamic (communication channel). Any simpler structure lacks one of these three necessary properties. Any more complex structure adds degrees of freedom that carry no new information.

The uniqueness of S^2 as the base is a theorem, not an assertion. For the S^1 fiber to be the unique communication channel, the base surface must be simply connected — otherwise the base itself carries non-contractible loops that compete with the

fiber as independent communication channels, producing unobserved additional circuit types. The classification of closed orientable surfaces is complete: they are enumerated by genus $g = 0, 1, 2, \dots$, and only genus $g = 0$ is simply connected. The torus T^2 (genus 1) has $\pi_1(T^2) = \mathbb{Z}^2$ and $H_1(T^2) = \mathbb{Z}^2$ — two independent non-contractible loops that would generate two additional families of circuits with no Standard Model counterpart. Every surface of genus $g \geq 1$ fails for the same reason. S^2 is the unique closed orientable surface satisfying the minimality requirement.

Orientability. Non-orientable surfaces (Klein bottle, projective plane $\mathbb{RP}^2$) are excluded by a separate requirement: the S^1 fiber must define a consistent handedness for circuit winding. On a non-orientable surface, the fiber direction reverses when transported around an orientation-reversing loop — making “winding number 1” and “winding number -1 ” circuits physically identical. The resulting absence of a conserved winding direction eliminates the distinction between circuits and anti-circuits, predicting no conserved electromagnetic charge. The substrate base must be orientable. Together with the genus-0 requirement, this uniquely selects S^2 .

2.3 Three Dimensions from Minimality

Three spatial dimensions emerge because three is the minimum dimensionality of a self-communicating surface: two for the surface (S^2), one for the fiber (S^1). No extra dimensions are required or predicted. This is BST’s answer to “why three dimensions” — three is the unique answer to “what is the minimum dimensionality of a self-organizing information surface.”

2.4 Structural Elements

Four structural elements: one surface (S^2), one fiber (S^1), one operation (contact commitment), one constraint (Haldane exclusion with capacity 137). Everything that follows in the remaining 25 sections is a consequence of these four elements and their geometry.

Section 3: The Contact Graph

3.1 Bubbles, Contacts, and Phases

The fundamental objects in BST are bubbles — featureless entities whose only property is whether they are in contact with other bubbles. The contact pattern defines a graph: nodes are bubbles, edges are contacts. Each contact carries an S^1 phase encoding the relationship between the two bubbles.

The contact graph is not embedded in a pre-existing space. Space emerges from the contact graph through holonomy — the pattern of S^1 phases around closed loops on the graph encodes the curvature of the emergent three-dimensional geometry (Section 18).

3.2 Circuits

A circuit is a closed path on the contact graph — a sequence of contacts that returns to its starting point. Circuits are the fundamental dynamical objects. A circuit's winding number on S^1 — the total phase accumulated around the loop — is an integer. This integer is topologically protected: small perturbations cannot change a winding number. Integer winding numbers are stable.

Particles are stable circuits. The winding number is the charge. The topological protection is the reason particles are stable — an electron doesn't decay because its winding number is an integer and integers can't be changed continuously.

3.3 Channel Capacity

The S^1 fiber has finite capacity. The maximum number of non-overlapping circuits on S^1 is determined by the packing geometry of the configuration space. This maximum is the channel capacity: 137.

At any point on the substrate, at most 137 circuits can coexist without mutual interference. This is neither fermionic exclusion (maximum 1) nor bosonic freedom (unlimited). It is Haldane fractional exclusion statistics with parameter $g = 1/137$.

Section 4: From Contact Graph to Configuration Space

4.1 Counting the Contact Degrees of Freedom

Before establishing the CR structure, we count the independent complex degrees of freedom in the BST contact geometry. This counting determines the dimension of the configuration space and motivates all that follows.

The gauge structure of BST has two sectors, each with a definite number of complex contact degrees of freedom:

Color sector ($SU(3)$): Quark circuits close on $\mathbb{CP}^2$, the configuration space for Z_3 -complete color triads. $\mathbb{CP}^2$ has complex dimension 2, but the Z_3 closure constraint adds one additional independent complex degree — the relative phase between the three color orderings. This gives $N_c = 3$ complex dimensions.

Electroweak sector ($SU(2) \times U(1)$): The Hopf fibration $S^3 \rightarrow S^2$ mediates the electroweak interaction. The base $S^2 \cong \mathbb{CP}^1$ has complex dimension 1. The S^1 communication fiber adds one more. This gives $N_w = 2$ complex dimensions.

The total number of independent complex contact degrees of freedom is:

$$\dim_{\mathbb{C}} = N_c + N_w = 3 + 2 = 5$$

This is not a coincidence. It is the reason the relevant bounded symmetric domain has complex dimension 5, and it ties the gauge structure of the Standard Model directly to the geometry of the configuration space.

4.2 The CR Structure

The BST contact manifold — the space of bubble contacts with their S^1 phase relationships — carries a natural Cauchy-Riemann (CR) structure. The CR structure arises because the contact geometry has both a real component (which bubbles touch) and a complex component (the S^1 phase at each contact). The CR manifold is strictly pseudoconvex because the contact form is non-degenerate.

The contact form. The natural candidate for the BST contact form is

$$\alpha = d\theta_f - A_{\text{Berry}}$$

where $\theta_f \in S^1$ is the fiber phase and A_{Berry} is the Berry connection 1-form accumulated as a contact moves over S^2 . The Berry connection on S^2 is precisely the Hopf connection of the fibration $S^1 \rightarrow S^3 \rightarrow S^2$ — the same structure already central to the electroweak sector in BST. The contact condition $\alpha \wedge (d\alpha)^5 \neq 0$ and strict pseudoconvexity of the Levi form $L(X, Y) = d\alpha(X, JY) > 0$ follow from the non-degeneracy of the Hopf connection. Explicit verification in local coordinates is identified as a near-term task; the algebraic shadow of this structure — the complex structure J on the tangent space $\mathfrak{m}$ confirmed in the Lie algebra verification (Section 4.4) — is already established.

4.3 The Derivation Chain

Notation. Throughout this paper, $\text{SO}_0(5, 2)$ denotes the connected component of the identity in the real Lie group $\text{SO}(5, 2)$ — the *non-compact* real form of the complex Lie algebra $\mathfrak{so}(7, \mathbb{C})$. Its Lie algebra $\mathfrak{so}(5, 2)$ has dimension $21 = 7 \times 6/2$. The *compact* real form of the same complex algebra is $\text{SO}(7)$, also dimension 21 — both groups have the same underlying vector space structure but different signatures. In mode-counting arguments (e.g., $T_c = N_{\text{max}} \times 20/21$), the 21 generators refer to $\dim \mathfrak{so}(5, 2)$, not to $\text{SO}(7)$. The isotropy subgroup $K = \text{SO}(5) \times \text{SO}(2)$ is compact.

The derivation from BST substrate geometry to the bounded symmetric domain D_{IV}^5 proceeds through a sequence of established mathematical theorems.

Step I: CR manifold identification. The contact manifold of BST bubble space is a strictly pseudoconvex CR manifold of complex dimension 5. The five complex dimensions are precisely the $N_c + N_w = 3 + 2$ contact degrees of freedom counted in Section 4.1.

Step II: Automorphism group. By the theorem of Chern and Moser (1974), the automorphism group of a strictly pseudoconvex CR manifold of complex dimension n is a subgroup of $\text{SU}(n+1, 1)$. For $n = 5$, this gives $\text{SU}(6, 1)$. The BST contact structure's additional real symmetry — the S^1 fiber is real-analytic with a real structure (complex conjugation) preserving the contact form — restricts the automorphism group from $\text{SU}(6, 1)$ to a real form of $\text{SO}(7, \mathbb{C}) \subset \text{SL}(7, \mathbb{C})$. The real forms of $\mathfrak{so}(7, \mathbb{C})$ are $\text{SO}(7)$, $\text{SO}(6, 1)$, $\text{SO}(5, 2)$, and $\text{SO}(4, 3)$. Of these, $\text{SO}(5, 2)$ is uniquely selected by two requirements: (1) non-compactness (the domain must be unbounded for

physics), which excludes $SO(7)$; and (2) Hermitian symmetry (Step III requires a Hermitian symmetric space for the Harish-Chandra bounded domain embedding and the Bergman kernel construction). A symmetric space G/K is Hermitian if and only if the center of K contains a circle group $U(1)$ (Helgason 1978, Ch. X, Theorem 6.1). Checking: $SO(6, 1)$ has $K = SO(6)$ with discrete center $\mathbb{Z}_2$ — not Hermitian; $SO(4, 3)$ has $K = SO(4) \times SO(3)$ with finite center — not Hermitian; $SO(5, 2)$ has $K = SO(5) \times SO(2)$ where the $SO(2) \cong U(1)$ factor furnishes the complex structure J on the tangent space $\mathfrak{m}$ — Hermitian. Therefore $SO(5, 2)$ is the unique non-compact real form of $so(7, \mathbb{C})$ whose associated symmetric space is Hermitian. This classification fact is confirmed by the explicit construction in Section 4.4, where the complex structure $J^2 = -1$ on $\mathfrak{m} \cong \mathbb{C}^5$ is verified computationally.

Step III: Harish-Chandra embedding. By Harish-Chandra's theorem (1956), every Hermitian symmetric space of non-compact type admits a canonical embedding as a bounded symmetric domain. The Hermitian symmetric space associated with $SO(5, 2)$ is the type IV domain D_{IV}^5 in Cartan's classification.

Step IV: Domain identification. Hua's classification of bounded symmetric domains (1958) establishes that D_{IV}^5 is a five-complex-dimensional domain with Shilov boundary isomorphic to $SO(5) \times SO(2)/SO(5)$. The Bergman kernel on this domain is known explicitly.

Step V: Physical interpretation. D_{IV}^5 is the configuration space of the BST contact graph. Points in the domain correspond to states of the contact graph. The Bergman metric on the domain determines the natural distance between configurations. The Shilov boundary determines the extremal configurations — the maximum-packing states.

Structural consequence: root system confirms $N_c = 3$. The identification of D_{IV}^5 carries an immediate algebraic corollary. The restricted root system of the symmetric space $D_{IV}^n = SO_0(n, 2)/[SO(n) \times SO(2)]$ is of type BC_2 , with long root multiplicity $a = 1$ and short root multiplicity $b = n - 2$ (Helgason 1978). For $n = n_C = 5$:

$$b = n_C - 2 = 3 = N_c$$

The short root multiplicity of the restricted root system of D_{IV}^5 is exactly the number of quark colors. Section 6.4 establishes $N_c = 3$ from the topological Z_3 closure requirement on quark triads. The root system of the domain establishes the same number algebraically: $b = n_C - 2 = 3$. Both derivations trace back to the single constraint $n_C = 5$ forced by the channel capacity formula $C = 137$. The same short root multiplicity appears in the Weyl vector table of Section 5.1, where it determines the $\rho_2 = (n_C - 2)/2$ component that fixes the numerator of the Wyler formula. The consistency — topology, root system, and channel capacity all yielding $N_c = 3$ — is a structural check on the D_{IV}^5 identification at the representation-theoretic level.

4.4 Critical Technical Point

The derivation chain requires that the local isotropy group of the BST contact structure is exactly $SO(5) \times SO(2)$. If the isotropy group contained additional discrete factors, the automorphism group could be smaller than $SO(5, 2)$, and the identification with D_{IV}^5 would fail. This has been verified by explicit Lie algebra construction. Seven independent checks all pass (see below).

Lie Algebra Verification (March 2026). The claim has been checked by explicit construction. Using 7×7 matrix representatives of $\mathfrak{so}(5, 2)$, the Cartan decomposition $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{m}$ with $\mathfrak{k} = \mathfrak{so}(5) \oplus \mathfrak{so}(2)$ was verified computationally:

Check	Pairs tested	Result
All generators in $\mathfrak{so}(5, 2)$	21	✓
$[\mathfrak{k}, \mathfrak{k}] \subseteq \mathfrak{k}$	121	✓
$[\mathfrak{k}, \mathfrak{m}] \subseteq \mathfrak{m}$	110	✓
$[\mathfrak{m}, \mathfrak{m}] \subseteq \mathfrak{k}$ — symmetric space	100	✓
Dimension chain: $\dim_{\mathbb{C}}(G/K) = 5$, Shilov = $S^4 \times S^1$	—	✓
$SO(2)$ rotation of fiber pairs	2	✓
Complex structure $J^2 = -1$ on $\mathfrak{m} \cong \mathbb{C}^5$	10	✓

All 7 checks pass. The symmetric space condition $[\mathfrak{m}, \mathfrak{m}] \subseteq \mathfrak{k}$ — the defining algebraic property of D_{IV}^5 — holds exactly. The complex structure on $\mathfrak{m}$ confirms that D_{IV}^5 is a Hermitian symmetric space, which is the condition required for the Bergman kernel construction to produce Wyler's formula.

Full details, generator conventions, and reproducible Python code are in the companion document `LieAlgebraVerification.md`.